Inequalities (😞≤🙂)

CPMSoc Mathematics

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Outline

Contents

Why Do We Care About Inequalities

- They are often hard to prove, so they appear in settings such as Extension 2 Mathematics in the NSW syllabus or mathematics olympiads.
- They are important to a field of mathematics called **real analysis**, which uses inequalities for:
  - Giving upper and lower bounds (using $\leq$ and $\geq$ inequalities) on the number of mathematical objects or quantities. This also relates to **graph theory** and **enumerative combinatorics**.
  - They are extremely important for notions of **convergence** and **limits**, where objects and quantities approach others in some sense.

Attendance

Please scan the QR code; and let us know about any dietary requirements.

(For the pizza!)

Trivial Inequality

$$\text{If } x \text{ is a real number, then } x^2 \geq 0.$$

This can be used for an advanced technique called Sum-Of-Squares (SOS), we will not cover in this workshop.

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Why is it called that? More generally, this also works for n terms.

Triangle Inequality

For $a_1, ..., a_n \in \mathbb{C}$, we have that

$$\left|\sum_{i=1}^{n} a_i\right| \leq \sum_{i=1}^{n} |a_i|.$$

General versions of Triangle Inequality

Triangle for vectors

For vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}^n$*, we have that*

$$|\boldsymbol{a} + \boldsymbol{b}| \leq |\boldsymbol{a}| + |\boldsymbol{b}|$$

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When using p-norms or L^p-space norms, the above inequality is typically called Minkowski's Inequality. Don't worry if you don't know this, we will mainly focus on the non-norm version.

Example Question

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Solution. Let $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ denote the sides with length $\log_{10} 12, \log_{10} 75, \log_{10} n$ respectively. Then by the triangle inequality,

$$|\boldsymbol{c}| = |\boldsymbol{a} - \boldsymbol{b}| < |\boldsymbol{a}| + |\boldsymbol{b}|,$$

so we obtain that

$$\log_{10} n < \log_{10} 12 + \log_{10} 75$$
$$\implies n < 900.$$

Similarly, choosing a different pair of sides, we obtain that $n > 6.25$, hence there could be 893 possible triangles.

Cauchy-Schwarz Inequality

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For $a_1, \dots, a_n, b_1, \dots, b_n \in \mathbb{R}$*, we have that*

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \leq \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right).$$

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If you've seen linear algebra, this form of Cauchy-Schwarz might also be familiar.

Cauchy-Schwarz Inequality in Inner Product Spaces

For vectors $\boldsymbol{u}, \boldsymbol{v}$ in a vector space with inner product $\langle \cdot, \cdot \rangle$, we have that

$$|\langle \boldsymbol{u}, \boldsymbol{v} \rangle| \leq \|\boldsymbol{u}\| \|\boldsymbol{v}\|,$$

where $\|\boldsymbol{u}\| = \langle \boldsymbol{u}, \boldsymbol{u} \rangle$.

Other Forms of Cauchy-Schwarz

Engel Form of Cauchy's Form

For real numbers $a_1, \ldots a_n$ *and positive real numbers* $b_1, \ldots, b_n$*, we have that*

$$\sum_{i=1}^{n} \frac{a_i^2}{b_i} \geq \frac{\left(\sum_{i=1}^{n} a_i\right)^2}{\sum_{i=1}^{n} b_i}.$$

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$$\sum_{i=1}^{n} \frac{a_i^2}{b_i} \geq \frac{\left(\sum_{i=1}^{n} a_i\right)^2}{\sum_{i=1}^{n} b_i}.$$

Proof: Since $b_i > 0$ for all $i \in \{1, \ldots, n\}$, we have that

$$\left(\sum_{i=1}^{n} a_i\right)^2 = \left(\sum_{i=1}^{n} \sqrt{b_i}\frac{a_i}{\sqrt{b_i}}\right)^2 \leq \left(\sum_{i=1}^{n} b_i\right)\left(\sum_{i=1}^{n} \frac{a_i^2}{b_i}\right),$$

and by rearranging we obtain the desired result! □

Example Question

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Let $a_1, \ldots, a_n, b_1, \ldots, b_n$ *be positive real numbers such that* $a_1 + \cdots + a_n = b_1 + \cdots + b_n$. *Prove that*

$$\frac{a_1^2}{a_1+b_1} + \cdots + \frac{a_n^2}{a_n+b_n} \geq \frac{a_1+\cdots+a_n}{2}.$$

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Solution. Using the Engel Form of Cauchy-Schwarz, set $x_i = a_i, y_i = a_i + b_i$, then

$$\sum_{i=1}^{n} \frac{x_i^2}{y_i} \geq \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{\sum_{i=1}^{n} y_i}.$$

$$\Longrightarrow \sum_{i=1}^{n} \frac{a_i^2}{a_i+b_i} \geq \frac{\left(\sum_{i=1}^{n} a_i\right)^2}{\sum_{i=1}^{n}(a_i+b_i)} = \frac{\left(\sum_{i=1}^{n} a_i\right)^2}{2\sum_{i=1}^{n} a_i} = \frac{\sum_{i=1}^{n} a_i}{2}$$

and we are done!

AM-GM Inequality

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AM-GM Inequality

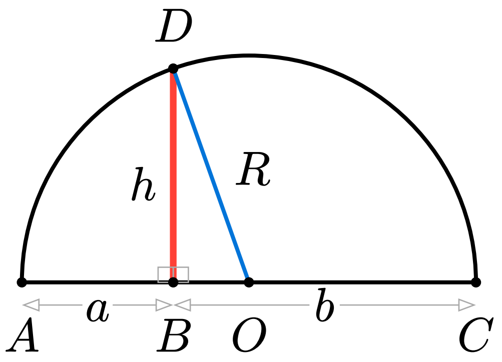
For any real numbers $x_1, x_2, \ldots, x_n \geq 0$,

$$\frac{x_1 + x_2 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \cdots x_n}$$

with equality if and only if $x_1 = x_2 = \ldots = x_n$.

AM-GM

For 2 variables, there is a neat visual proof for the AM-GM inequality.
In the diagram the radius is $R = OC = OD = \frac{a+b}{2}$.



The distance OB is the radius minus segment b, $OB = OC - BC = \frac{a+b}{2} - a = \frac{b-a}{2}$. Applying the Pythagorean theorem to the $\triangle OBD$ gives $\left(\frac{b-a}{2}\right)^2 + h^2 = \left(\frac{a+b}{2}\right)^2$.
Solving for h yields $h^2 = \left(\frac{a+b}{2}\right)^2 - \left(\frac{b-a}{2}\right)^2 = ab$.
So $h = \sqrt{ab}$.

Since a leg ($BD = h$) cannot be longer than the hypotenuse ($OD = R$), $h \leq R \implies \sqrt{ab} \leq \frac{a+b}{2}$. Equality holds if and only if B coincides with O ($a = b$).

AM-GM Proof

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- **Base case**: Prove the inequality for 2 variables
- **Forward step:** If the inequality holds for n variables, then it holds for $2n$ variables.
- **Backward step:** If the inequality holds for n variables, then it holds for $n - 1$ variables.

(Think about why this works!)

AM-GM Proof

Base case ($n = 2$):

Expand $\left(\sqrt{a} - \sqrt{b}\right)^2 = a + b - 2\sqrt{ab} \geq 0 \implies \frac{a+b}{2} \geq \sqrt{ab}$.

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Forward step ($n \to 2n$): Let $A = \frac{a_1 + \ldots + a_n}{n}$ and $B = \frac{a_{n+1} + \ldots + a_{2n}}{n}$.
By n-variable AM-GM, $A \geq \sqrt[n]{a_1 \ldots a_n}$ and $B \geq \sqrt[n]{a_{n+1} \ldots a_{2n}}$. Then,

$$\frac{a_1 + \ldots + a_{2n}}{2n} = \frac{A + B}{2} \geq \sqrt{AB} \geq \sqrt{\sqrt[n]{a_1 \ldots a_n} \cdot \sqrt[n]{a_{n+1} \ldots a_{2n}}} = \sqrt[2n]{a_1 \ldots a_{2n}}.$$

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$$\frac{a_1 + \ldots + a_{2n}}{2n} = \frac{A + B}{2} \geq \sqrt{AB} \geq \sqrt{\sqrt[n]{a_1 \ldots a_n} \cdot \sqrt[n]{a_{n+1} \ldots a_{2n}}} = \sqrt[2n]{a_1 \ldots a_{2n}}.$$

Backward step ($n \to n - 1$):

Let $a_n = \frac{a_1 + \ldots + a_{n-1}}{n-1}$. Then $\frac{a_1 + \ldots + a_n}{n} = A$. By n-variable AM-GM:

$$\frac{a_1 + \ldots + a_{n-1} + \frac{a_1 + \ldots + a_{n-1}}{n-1}}{n} = \frac{a_1 + \ldots + a_{n-1}}{n-1}.$$

So

$$\frac{a_1 + \ldots + a_{n-1}}{n-1} \geq \sqrt[n]{a_1 a_2 \ldots a_{n-1} \frac{a_1 + \ldots + a_{n-1}}{n-1}}$$

$$\left(\frac{a_1 + \ldots + a_{n-1}}{n-1}\right)^{\frac{n-1}{n}} \geq \sqrt[n]{a_1 a_2 \ldots a_{n-1}}.$$

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Solution: The denominators on LHS are certainly very annoying, so let's get rid of them!

By using the 3-term AM-GM on the terms below, we get that

$$\frac{a^3}{bc} + b + c \geq 3a, \quad \frac{b^3}{ca} + c + a \geq 3b, \quad \frac{c^3}{ab} + a + b \geq 3c.$$

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Summing the above statements gives us:

$$\frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab} + 2(a+b+c) \geq 3(a+b+c).$$

Which rearranges to our desired result.

Jensen’s Inequality

Convexity - Prelude to Jensen's Inequality

Definition

Convexity refers to the property of a function having a secant lying above the curve. For a convex function f, a point c on $[x_1, x_2]$ where $x_2 \geq x_1$ would have $f(c) \geq f(x_1), f(x_2)$.

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Property of Convex Function

Convex functions also have $f''(x) \geq 0$. Remember not twice differentiable functions can be convex too!

Concavity - Prelude to Jensen's Inequality

Definition

Opposite of being convex, so a secant below the curve. For a concave function f, $c \in [x_1, x_2]$ *where* $x_2 \geq x_1$ *would demonstrate* $f(c) \leq f(x_1), f(x_2)$.

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Jensen's Inequality

For a convex function f (opposite true for concave)

$$f\left(\sum_{i=1}^{n} t_i x_i\right) \leq \sum_{i=1}^{n} t_i f(x_i), \qquad \sum_{i=1}^{n} t_i = 1$$

Jensen's Inequality Proof

Proof: We proved by induction

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$n = 2$ (BASE CASE)

Let $t_1 + t_2 = 1$ with $t_i \geq 0$, s.t. $t_2 = 1 - t_1$.

Then any point in the interval $[x_1, x_2]$ can be represented as $x = t_1 x_1 + t_2 x_2$.

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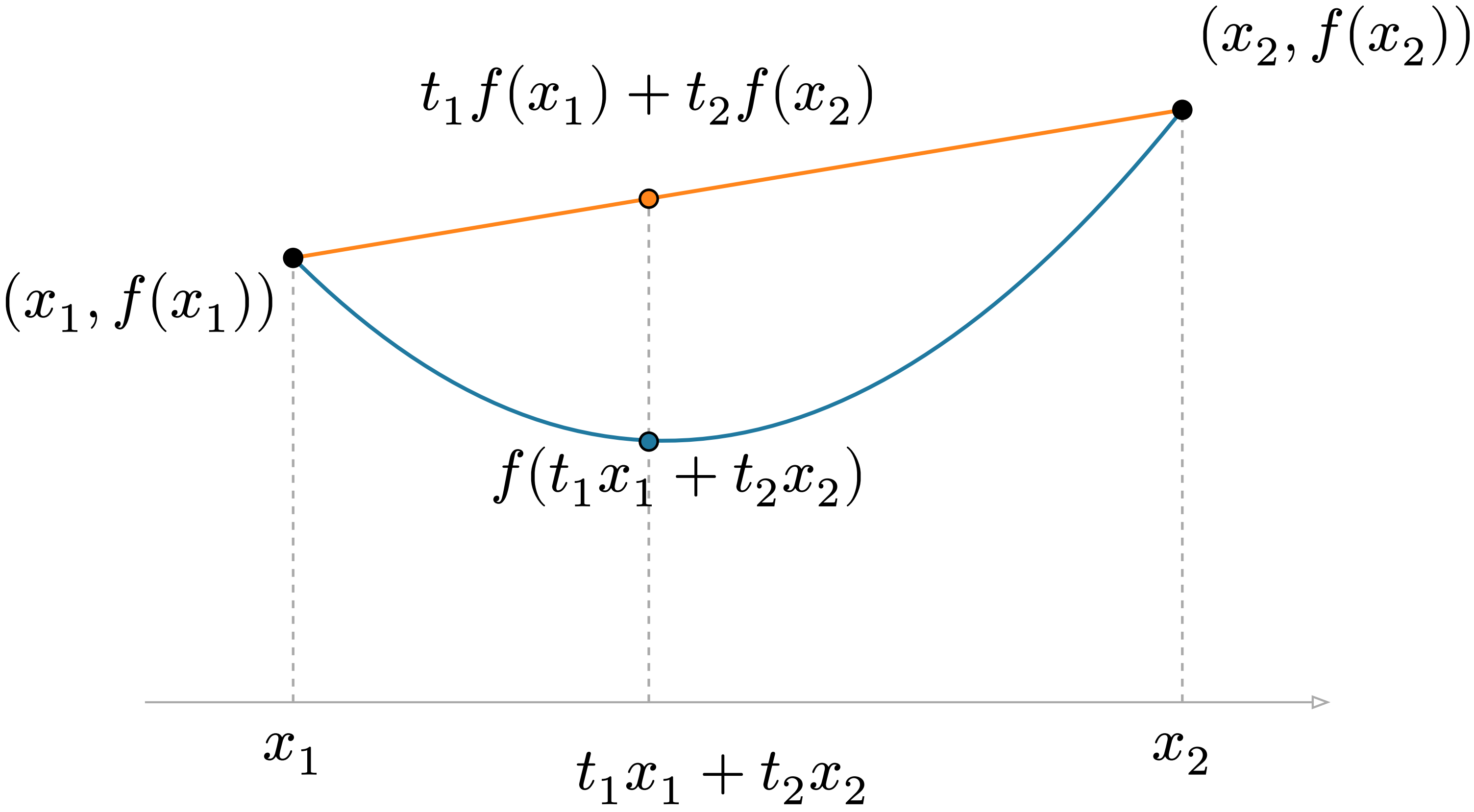
Then any point in the interval $[x_1, x_2]$ can be represented as $x = t_1 x_1 + t_2 x_2$.

A function f is convex by definition if the chord connecting $(x_1, f(x_1))$ and $(x_2, f(x_2))$ lies on or above the curve: $f(t_1 x_1 + t_2 x_2) \leq t_1 f(x_1) + t_2 f(x_2)$ making the base case true.

Jensen's Inequality Proof



Jensen's Inequality Proof

$n = k$ (GENERAL CASE)

Assume Jensen's Inequality holds for $n = k$, then

$$f\left(\sum_{i=1}^{k} t_i x_i\right) \leq \sum_{i=1}^{k} t_i f(x_i). \tag{1}$$

We are required to prove given Jensen's Inequality holds for $n = k$, it also holds for $n = k + 1$, which is

$$f\left(\sum_{i=1}^{k+1} t_i x_i\right) \leq \sum_{i=1}^{k+1} t_i f(x_i). \tag{2}$$

Jensen's Inequality Proof

LHS of (2) $= f\left(\sum_{i=1}^{k+1} t_i x_i\right) = f\left(t_{k+1}x_{k+1} + (1 - t_{k+1}) \sum_{i=1}^{k} \frac{t_i}{1-t_{k+1}} x_i\right).$

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Applying Jensen's inequality for $n = 2$,

LHS of $(2) \ \leq t_{k+1} f(x_{k+1}) + (1 - t_{k+1}) f\left(\sum_{i=1}^{k} \frac{t_i}{1-t_{k+1}} x_i\right)$

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Since $t_1 + \ldots + t_{k+1} = 1$, $\sum_{i=1}^{k} t_i = (1 - t_{k+1})$ under (2) s.t. $\sum_{i=1}^{k} \frac{t_i}{1-t_{k+1}} = 1$.
This means we can apply out inductive hypothesis to this expression.

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This means we can apply out inductive hypothesis to this expression.

LHS of (2) $\leq \ldots \leq \; t_{k+1} f(x_{k+1}) + (1 - t_{k+1}) \sum_{i=1}^{k} \frac{t_i}{1-t_{k+1}} f(x_i)$

$= \sum_{i=1}^{k} t_i f(x_i) + t_{k+1} f(x_{k+1}) = \sum_{i=1}^{k+1} t_i f(x_i) =$ RHS of (2).

$\square$

Jensen's Inequality Simple Application

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By Jensen's Inequality, such a concave function would meet the following:

$$f(t_1 x_1 + \ldots + t_4 x_4) \geq t_1 f(x_1) + \ldots + t_4 f(x_4)$$

Jensen's Inequality Simple Application

Substituting f, letting $(t_1, ..., t_4) = (b, c, d, e), (x_1, ..., x_4) = (a, b, c, d)$ where our substitution for t's is valid as $b + c + d + e = 1$.

$\ln(ba + cb + dc + ed) \geq b\ln(a) + c\ln(b) + d\ln(c) + e\ln(d)$.

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$\ln(ba + cb + dc + ed) \geq b \ln(a) + c \ln(b) + d \ln(c) + e \ln(d)$.

After log laws and rearranging, $\ln(ab + bc + cd + de) \geq \ln\left(a^b b^c c^d d^e\right)$.

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After log laws and rearranging, $\ln(ab + bc + cd + de) \geq \ln\left(a^b b^c c^d d^e\right)$.

Now since exponentiation is an increasing function, we can cancel out the log.

Hence, $ab + bc + cd + de \geq a^b b^c c^d d^e$ for $b + c + d + e = 1$ proven as required.

A Tasty Application of Jensen's Inequality :)

Example

IMO 2001:

Let a, b, c *be positive real numbers. Prove* $\frac{a}{\sqrt{a^2+8bc}} + \frac{b}{\sqrt{b^2+8ac}} + \frac{c}{\sqrt{c^2+8ab}} \geq 1$

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This problem has multiple approaches such as with homogeneity that enables assuming $a + b + c = 1$ to later enable Jensen's and applying other inequalities such as Hölder. We will consider an algebra heavy Jensen's method.

However, there is a more algebra heavy method relying on algebraic manipulations.

A Tasty Application of Jensen's Inequality :)

Firstly, note that we can express the left hand side of this inequality (LHS) as a scaled version of a 'weighted mean' of sort as we are positing the ansatz Jensen's inequality can be applied,

$$\frac{a}{\sqrt{a^2+8bc}}+\frac{b}{\sqrt{b^2+8ac}}+\frac{c}{\sqrt{c^2+8ab}} = 3\left(\frac{1}{3}\frac{a}{\sqrt{a^2+8bc}}+\frac{1}{3}\frac{b}{\sqrt{b^2+8ac}}+\frac{1}{3}\frac{c}{\sqrt{c^2+8ab}}\right).$$

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Now note that

$$\frac{a}{\sqrt{a^2+8bc}} = \frac{\sqrt{a^2}}{\sqrt{a^2+8bc}} = \frac{\sqrt{a^3}}{\sqrt{a^3+8abc}},$$

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and similarly

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such that we define a function $f(x) = \frac{\sqrt{x^3}}{\sqrt{x^3+8abc}}$ to express these terms as $f(a), f(b), f(c)$ respectively.

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$$f(x) = \left(1 + 8abcx^{-3}\right)^{-\frac{1}{2}},$$

$$f'(x) = -\tfrac{1}{2} \cdot -24abcx^{-4} \cdot \left(1 + 8abcx^{-3}\right)^{-\frac{3}{2}},$$

$$f'(x) = 12abc \cdot \left(x^{-4}\left(1 + 8abcx^{-3}\right)^{-\frac{3}{2}}\right),$$

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Consequently, $f''(x) \geq 0$ when $x \leq (abc)^{\frac{1}{3}}$ making f convex, we can apply Jensen's Inequality assuming $a, b, c \leq (abc)^{\frac{1}{3}}$.

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This means our expression simplifies to

$$3\left(\tfrac{1}{3}f(a) + \tfrac{1}{3}f(b) + \tfrac{1}{3}f(c)\right) \geq 3f\left(\tfrac{1}{3}(a+b+c)\right).$$

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$$\geq 3f\left((abc)^{\frac{1}{3}}\right)=3\left(\frac{\sqrt{abc}}{\sqrt{abc+8abc}}\right)=1 \text{ as required,}$$

under the assumption $a,b,c\leq(abc)^{\frac{1}{3}}$.

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under the assumption $a,b,c\leq(abc)^{\frac{1}{3}}$. This assumption does not limit the other approaches to this question we previously mentioned.

Hölder's Inequality

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For $a_1, \ldots, a_n, b_1, \ldots, b_n \in \mathbb{R}$*, and* $p, q \in \mathbb{R}$ *such that* $p, q \in [1, \infty]$ *and* $\frac{1}{p} + \frac{1}{q} = 1$*, we have that*

$$\sum_{i=1}^{n} |a_i b_i| \leq \left(\sum_{i=1}^{n} |a_i|^p\right)^{\frac{1}{p}} \left(\sum_{i=1}^{n} |b_i|^q\right)^{\frac{1}{q}}$$

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Or, more generally…

Hölder's Inequality

In a measure space (Ω, σ, μ), *if* f, g *are measurable functions with* $p, q \in [1, \infty]$ *and* $\frac{1}{p} + \frac{1}{q} = 1$, *then*

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Obviously, if you haven’t seen this, don’t worry about it!

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What happens when $p = 2, q = 2$?

Example Question

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Let x, y, k *be positive real numbers. Prove that*

$$\left(1+\frac{x}{y}\right)^k + \left(1+\frac{y}{x}\right)^k \geq 2^{k+1}.$$

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Observe that

$$\left(1+\frac{x}{y}\right)^k + \left(1+\frac{y}{x}\right)^k = \left(\frac{x+y}{y}\right)^k + \left(\frac{x+y}{x}\right)^k = (x+y)^k\left(x^{-k}+y^{-k}\right).$$

Example Continued

Let $p = \frac{k+1}{k}$, $q = k+1$, then clearly $\frac{1}{p} + \frac{1}{q} = 1$ and

$$\begin{aligned}(x+y)^{\frac{k}{k+1}}\left(x^{-k}+y^{-k}\right)^{\frac{1}{k+1}} &\geq x^{\frac{k}{k+1}}y^{-\frac{k}{k+1}} + x^{-\frac{k}{k+1}}y^{\frac{k}{k+1}} \\ &\geq 2\sqrt{x^{\frac{k}{k+1}}y^{-\frac{k}{k+1}}x^{-\frac{k}{k+1}}y^{\frac{k}{k+1}}} = 2,\end{aligned}$$

so by raising both sides to the power of $k+1$, we obtain the result. Umazing!

Homogenisation

Homogenisation

Motivating problem

Let $a, b, c > 0$ *and* $a + b + c = 1$. *Prove that*

$$a^2 + b^2 + c^2 + a + b + c \geq \frac{4}{3}.$$

Homogenisation

Motivating problem

Let $a, b, c > 0$ and $a + b + c = 1$. Prove that

$$a^2 + b^2 + c^2 + a + b + c \geq \frac{4}{3}.$$

This inequality mixes three different degrees at once,

$$a^2 + b^2 + c^2 \quad \text{(degree 2)}, \quad a + b + c \quad \text{(degree 1)}, \quad \frac{4}{3} \quad \text{(degree 0)}.$$

Homogenisation

Homogenisation means rewriting an inequality so all terms have the same degree.

Imagine replacing (a, b, c) by $(2a, 2b, 2c)$.

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Expression	**What happens when we double a, b, c**
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The original inequality is awkward to work with as its terms respond differently to the same change of scale. Homogenisation aims to fix that.

Homogenisation

To compare all terms fairly, make everything degree 2. As $a+b+c=1$,

$$\begin{aligned} a+b+c &\rightsquigarrow (a+b+c)^2, \\ \frac{4}{3} &\rightsquigarrow \frac{4}{3}(a+b+c)^2. \end{aligned}$$

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So it is enough to prove the homogeneous inequality

$$a^2+b^2+c^2+(a+b+c)^2 \geq \frac{4}{3}(a+b+c)^2,$$

$$\Leftrightarrow 3\left(a^2+b^2+c^2\right)-(a+b+c)^2=(a-b)^2+(b-c)^2+(c-a)^2 \geq 0.$$

Normalisation

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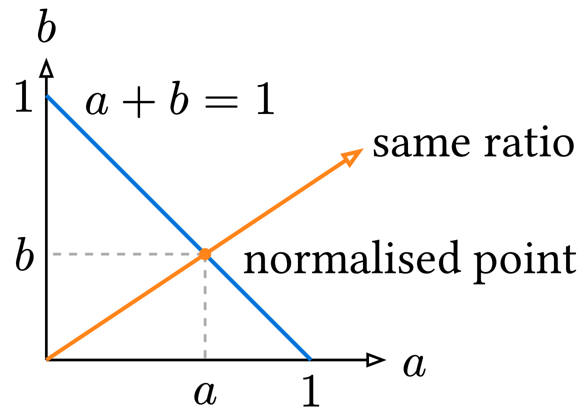
While **homogenisation** establishes scale independence, **normalisation** plays with it by choosing a specific, convenient scale for the problem.

Since a scale-invariant inequality depends only on the **ratio** of the variables, any positive scaling $t(a, b)$ preserves its truth value. So we can restrict our analysis to the unique representative of each ray that intersects a convenient constraint such as $a + b = 1$.

The choice of constraint is arbitrary, but we typically set the most frequent or algebraically complex expression to 1 to simplify calculations.



Normalisation

Problem

Let $a, b, c > 0$*. Prove that*

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{a+b+c}{2}.$$

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$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{a+b+c}{2}.$$

Both sides are homogeneous of degree 1, so we can normalise to $a+b+c=1$. This simplifies our target to

$$\frac{a^2}{1-a} + \frac{b^2}{1-b} + \frac{c^2}{1-c} \geq \frac{1}{2}.$$

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This is useful to us because the expression on the left is three of the same function, with only one variable, so Jensen's is directly applicable.

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Another useful property of normalisation applies here, since a, b, c sum to 1 and are positive, they must individually be less than 1. It is an exercise for the reader to prove that $f(x) = \dfrac{x^2}{1-x}$ is convex on $(-\infty, 1)$.

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$$\begin{aligned}\frac{a^2}{1-a} + \frac{b^2}{1-b} + \frac{c^2}{1-c} &= f(a) + f(b) + f(c) \\ &\geq 3f\left(\frac{a+b+c}{3}\right) = \frac{1}{2}.\end{aligned}$$

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Engle form of Cauchy Schwarz also works.

Some Useful Substitutions

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Roots Sub

If $a, b, c \geq 0$ and roots appear, set

$$a = x^2,$$
$$b = y^2, c = z^2.$$

We can obviously use this for other surds as well. This turns radicals into ordinary products.

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Example

For $a, b, c \geq 0$, prove

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$$a + b + c \geq \sqrt{ab} + \sqrt{bc} + \sqrt{ca}.$$

With the roots sub, the radicals become products, so it is enough to prove

$$x^2 + y^2 + z^2 \geq xy + yz + zx,$$

$$\Leftrightarrow x^2 + y^2 + z^2 - xy - yz - zx = \frac{1}{2} \sum_{\text{cyc}} (x - y)^2 \geq 0.$$

Some Useful Substitutions

Ratio Sub

Use ratios:

$$x = \frac{a}{b},$$

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Then $xyz = 1$. This is also useful when $xyz = 1$ is a constraint.

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Example

Prove, for $a, b, c > 0$,

$$\left(\frac{a}{b} + \frac{b}{c} + \frac{c}{a}\right)^2 \geq 3\left(\frac{a}{c} + \frac{b}{a} + \frac{c}{b}\right).$$

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With the ratio sub, $xyz = 1$ and $\frac{a}{c} = xy$, $\frac{b}{a} = yz$, $\frac{c}{b} = zx$. Thus it is enough to show

$$(x + y + z)^2 \geq 3(xy + yz + zx).$$

The proof follows from the previous slide.

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Ravi Sub

For triangle sides, write

$$a = y + z, \quad b = z + x, \\ c = x + y,$$

with $x, y, z > 0$.

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If a, b, c are sides of a triangle, prove

$$(a + b - c)(b + c - a)(c + a - b) \leq abc.$$

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$$(a + b - c)(b + c - a)(c + a - b) \leq abc.$$

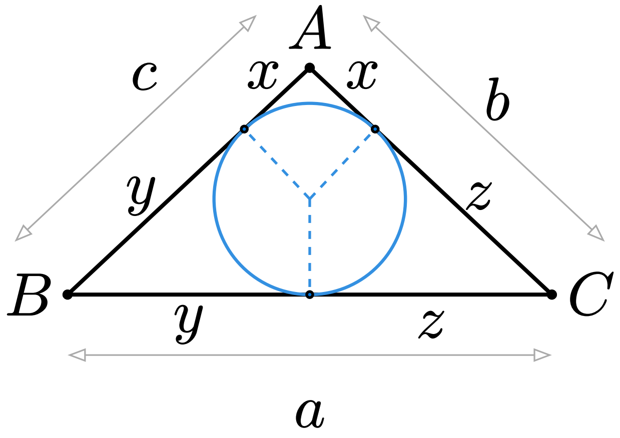
With the Ravi sub, $a + b - c = 2z, \quad b + c - a = 2x, \quad c + a - b = 2y$, so the left side is $8xyz$. And, by AM–GM,

$$\begin{aligned} abc &= (x + y)(y + z)(z + x) \\ &\geq 2\sqrt{xy} \cdot 2\sqrt{yz} \cdot 2\sqrt{zx} = 8xyz. \end{aligned}$$

Some Useful Substitutions

The idea behind the Ravi sub is that tangent segments from any vertex to the incircle (just think about it as the biggest circle we can make in a triangle) are equal. We can let these tangent lengths be x, y, and z, so

$a = y + z, \quad b = z + x \quad c = x + y.$



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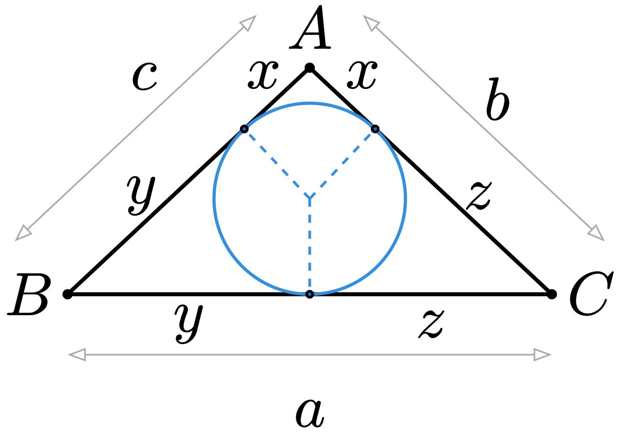

Substituting these expressions maps the triangle inequality onto the standard positive quadrant:

$$a + b > c \Leftrightarrow (y + z) + (z + x) > (x + y) \Leftrightarrow z > 0$$

So, any triangle inequality simply becomes $x, y, z > 0.$

Some Useful Substitutions

Trig Sub

Trig sub is useful when constraints resemble trig identities, and allows us to use Jensen's!

Consider $xy + yz + zx = 1$ for $x, y, z \geq 0$. We can let $x = \tan\left(\frac{A}{2}\right)$, $y = \tan\left(\frac{B}{2}\right)$, $z = \tan\left(\frac{C}{2}\right)$ for positive acute half-angles.

To be clear, $\sum$ *means sum of 3 elements* (A, B, C) *in this example.*

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With the trig sub, $\frac{x}{\sqrt{1+x^2}} = \sin\left(\frac{A}{2}\right)$, so the inequality becomes:

$$\sum \sin\left(\frac{A}{2}\right) \leq \frac{3}{2}.$$

Some Useful Substitutions

Now let's investigate what our original constraint turns into. Using the tangent addition formula:

$$\tan\left(\sum \frac{A}{2}\right) = \frac{\sum \tan\left(\frac{A}{2}\right) - \prod \tan\left(\frac{A}{2}\right)}{1-(xy+yz+zx)}.$$

We get $xy + yz + zx = 1$ precisely when the denominator is 0, meaning $\tan\left(\sum \frac{A}{2}\right) = \infty$. So $\sum \frac{A}{2} = \frac{\pi}{2}$, which yields $A + B + C = \pi$. It is satisfyingly constrained to a triangle!

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Since $f(u) = \sin\left(\frac{u}{2}\right)$ is concave on $[0, \pi]$, we apply Jensen's Inequality:

$$\begin{aligned} \sum \sin\left(\frac{A}{2}\right) &\leq 3\sin\left(\frac{A+B+C}{6}\right) \\ &= 3\sin\left(\frac{\pi}{6}\right) = \frac{3}{2}. \end{aligned}$$

Some Useful Substitutions

Thanks for coming! We hope you enjoyed 😋, pizza is coming.

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More mathematics workshops to come!!!

Feel free to try the worksheets and discuss!!!

Feedback Form